Keeping data private
in real-time pipelines

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Don’t expose them to bright light.

Don’t get them wet.

Never feed them after midnight.





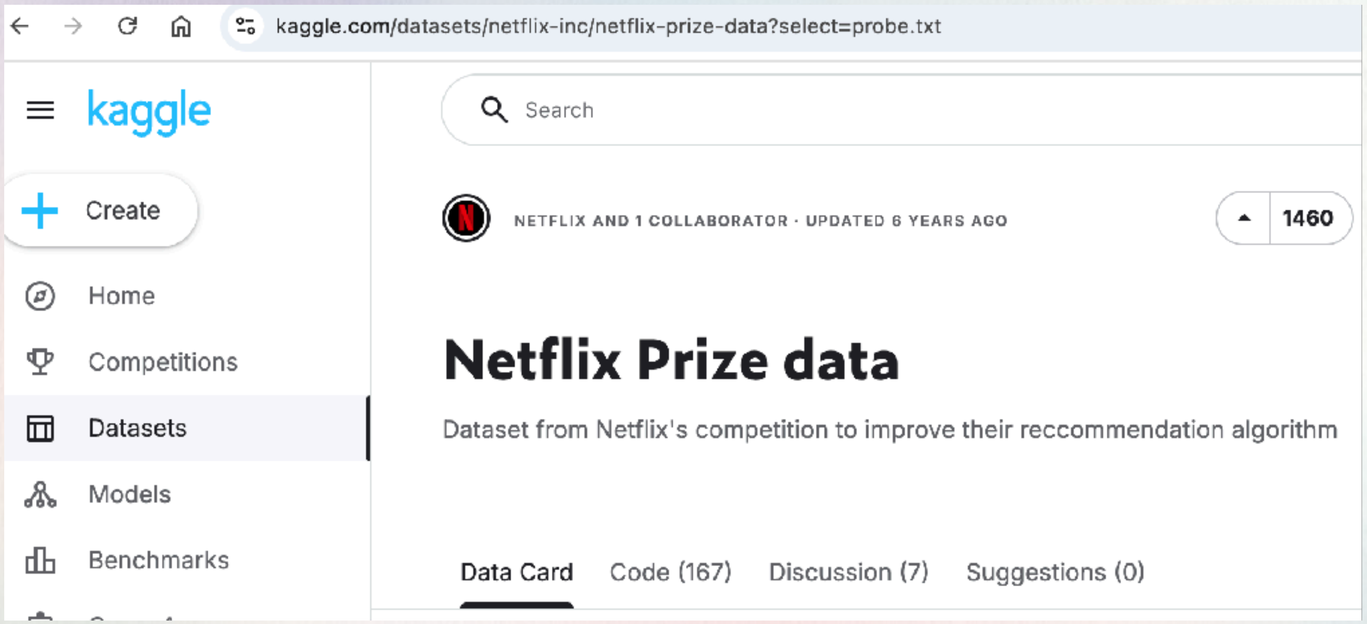
Privacy issues



Data



Netflix Prize Dataset





- a training data set of **100,480,507 ratings**

that **480,189 users** gave to **17,770 movies**

For each movie:
- CustomerID
- Rating
- Date

Robust De-anonymization of Large Datasets (How to Break Anonymity of the Netflix Prize Dataset)

Arvind Narayanan and Vitaly Shmatikov

The University of Texas at Austin

November 26, 2024

arXiv:cs/0610105v2 [cs.CR] 22 Nov 2007

Abstract

We present a new class of statistical de-anonymization attacks against high-dimensional micro-data, such as individual preferences, recommendations, transaction records and so on. Our techniques are robust to perturbation in the data and tolerate some mistakes in the adversary's background knowledge.

We apply our de-anonymization methodology to the Netflix Prize dataset, which contains anonymous movie ratings of 500,000 subscribers of Netflix, the world's largest online movie rental service. We demonstrate that an adversary who knows only a little bit about an individual subscriber can easily identify this subscriber's record in the dataset. Using the Internet Movie Database as the source of background knowledge, we successfully identified the Netflix records of known users, uncovering their apparent political preferences and other potentially sensitive information.

1 Introduction

Privacy nightmare in streaming systems



- Always on
- Low latency
- Automated
- No human checkpoint



produce

kafka

topic

copied

consumed







message



MongoDB.

```
{  "event_type": "TripRequested",
   "trip_id": "t-123",
   "rider_id": "r-456",
   "pickup": {"lat": 52.52, "lon": 13.40},
   "destination": {"lat": 52.50, "lon": 13.42},
   "timestamp": "2026-10-22T10:15:00Z"}
```

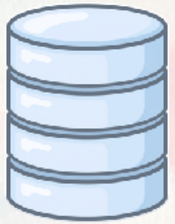
processed

enriched







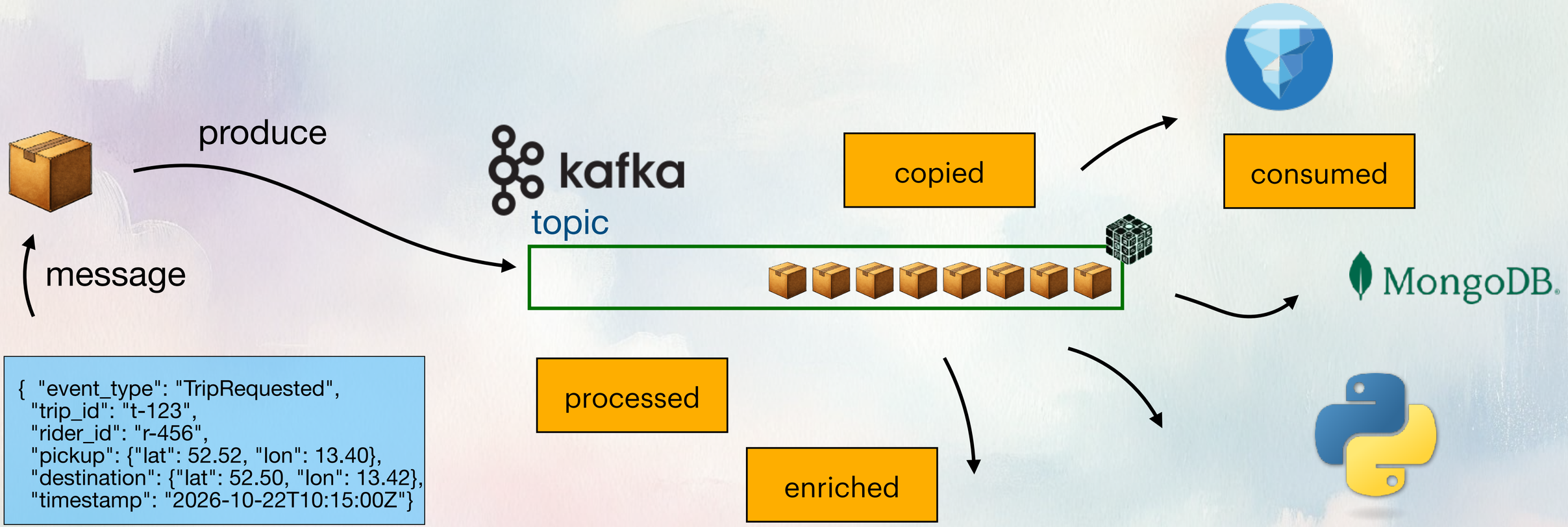




Level 1: foundation



Shift left



Protect sensitive data before it is copied, persisted, or fanned out.

Foundation



Identity & access control

- Authentication - who?
- Authorization - what?
- Role-based access control

Encryption



- At rest
- In transit
- Client-side field level encryption (CSFLE)
- Client-side payload encryption (CSPE)

Data governance & policy



- Schema Registry / data contracts
- PII tags
- Validation or transformation rules
- Lineage and catalog visibility

Audit logs and evidence for compliance

- Track user / application access
- Identify abnormal behavior
- Meet regulatory compliance requirements
- Investigate security incidents
- Monitor resource usage / operational changes



Level 2: data masking & tokenization



Data masking



- john.doe@gmail.com → j***@gmail.com
- Credit card → **** **** **** 1234
- Phone → +1-***-***-4421



Core principles

- Often retains the structure, format, and logic of the original
- Should prevent tracing back to the original values
- Be consistent when needed

Common masking techniques

- Substitution
- Shuffling
- Number/date variance
- Encryption
- Nulling out / deletion
- Masking out

Different types of data masking

- Static vs dynamic
- Deterministic or random
- Statistical or record-level
- On-the-fly or pre-computed

Tokenization



Email → tok_8f3a91
User ID → u_492001



How it works

- Tokenization replaces real values with meaningless stand-ins
- Tokens have no meaning outside the system
- Mapping lives in a secure store or service

Types of tokens

- High-Value Tokens - can be used in place of the real data
- Low-Value Tokens - only a pointer
- Deterministic and random

Key components

- Token generation
- Token mapping
- Token vault (data store)
- Key management

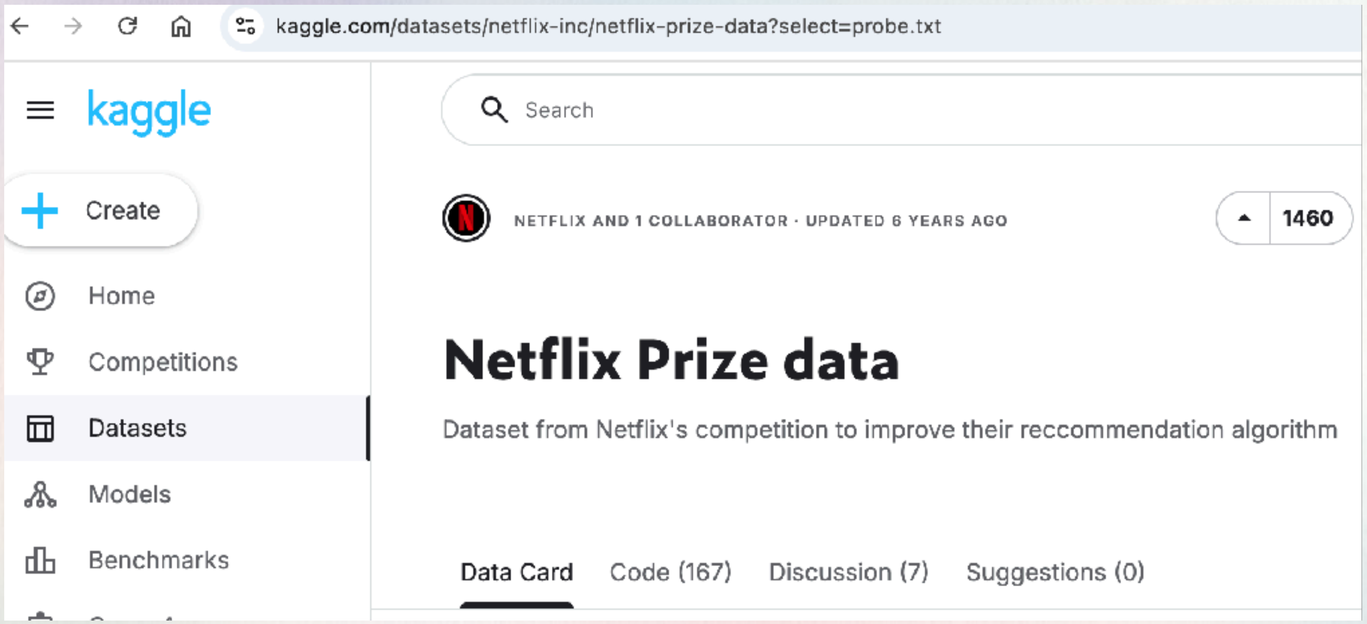
Tokenization vs encryption

- Tokenization replaces data with a surrogate
- Encryption mathematically transforms data
- Hashing is not tokenization

Re-identification attacks



Netflix Prize Dataset





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Medical records

https://techscience.org/a/2015092903/

techscience.org/a/2015092903/

Only You, Your Doctor, and Many Others May Know

Latanya Sweeney

Record	[illegible]
Hospital	162: Sacred Heart Medical Center in Providence
Admit Type	1: Emergency
Type of Stay	[illegible]
Length of Stay	6 days
Discharge Date	Oct-2011
Discharge Status	[illegible] under the care of an health service organization
Charges	$71708.47
Payers	1: Medicare 6: Commercial insurance 625: Other government sponsored patients
Emergency Codes	E8162: motor vehicle traffic accident due t[illegible] loss of control; loss control mv-mocycl
Diagnosis	[illegible] closed fracture

MAN, 60, THROWN FROM MOTORCYCLE

A 60-year-old Soap Lake man was hospitalized Saturday afternoon after he was thrown from his motorcycle. Ronald Jameson was riding his 2003 Harley-Davidson north on Highway 25, when he failed to negotiate a curve to the left. His motorcycle became airborne before landing in a wooded area. Jameson was thrown from the bike;

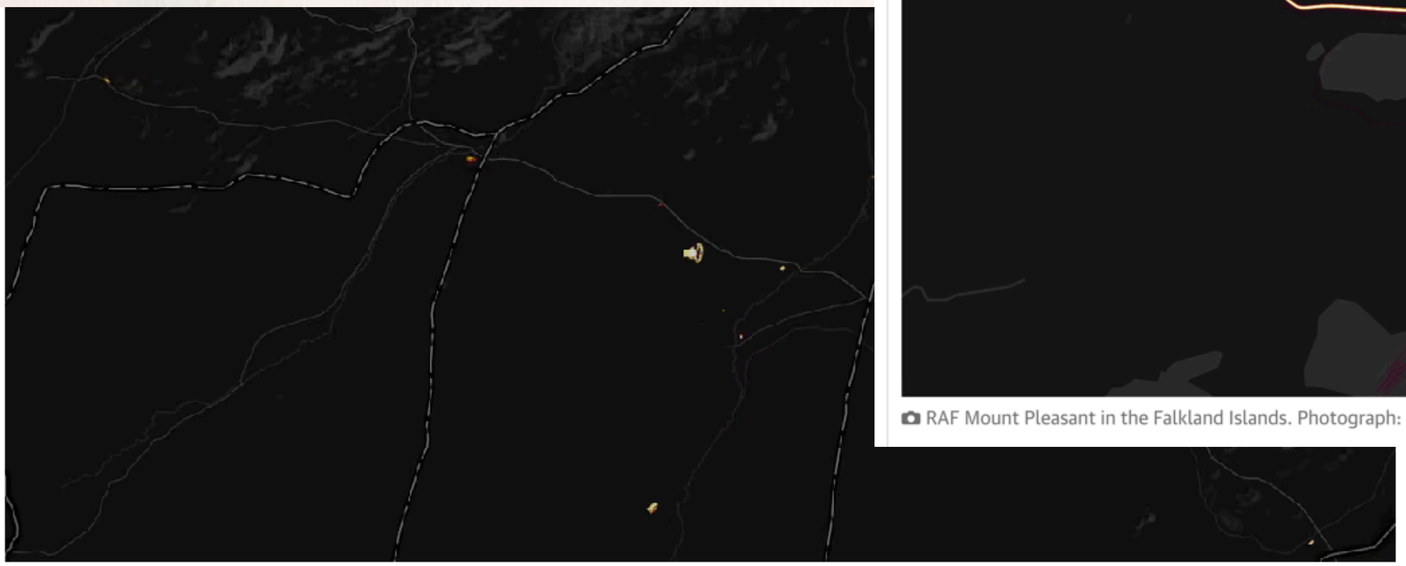
Strava tracking data

- every single activity ever uploaded to Strava – more than **3 trillion individual GPS data points**

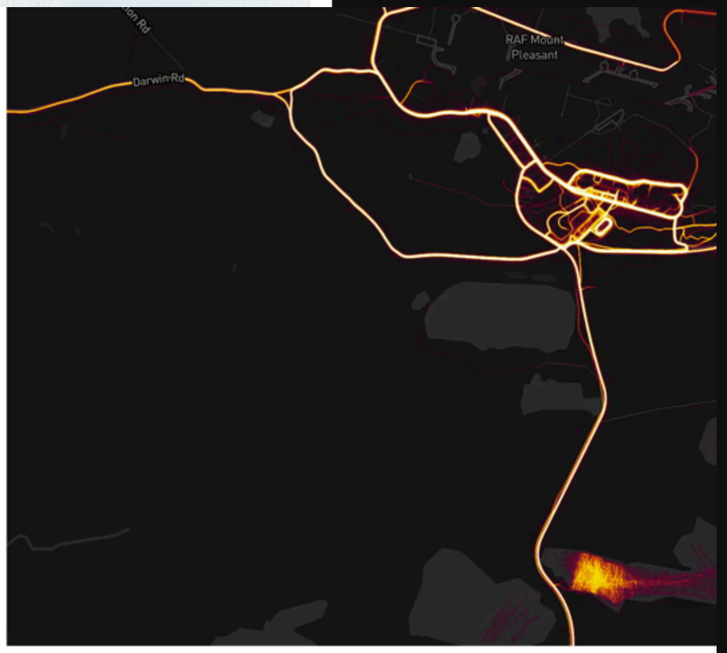
Fitness tracking app Strava gives away location of secret US army bases

Data about exercise routes shared online by soldiers can be used to pinpoint overseas facilities

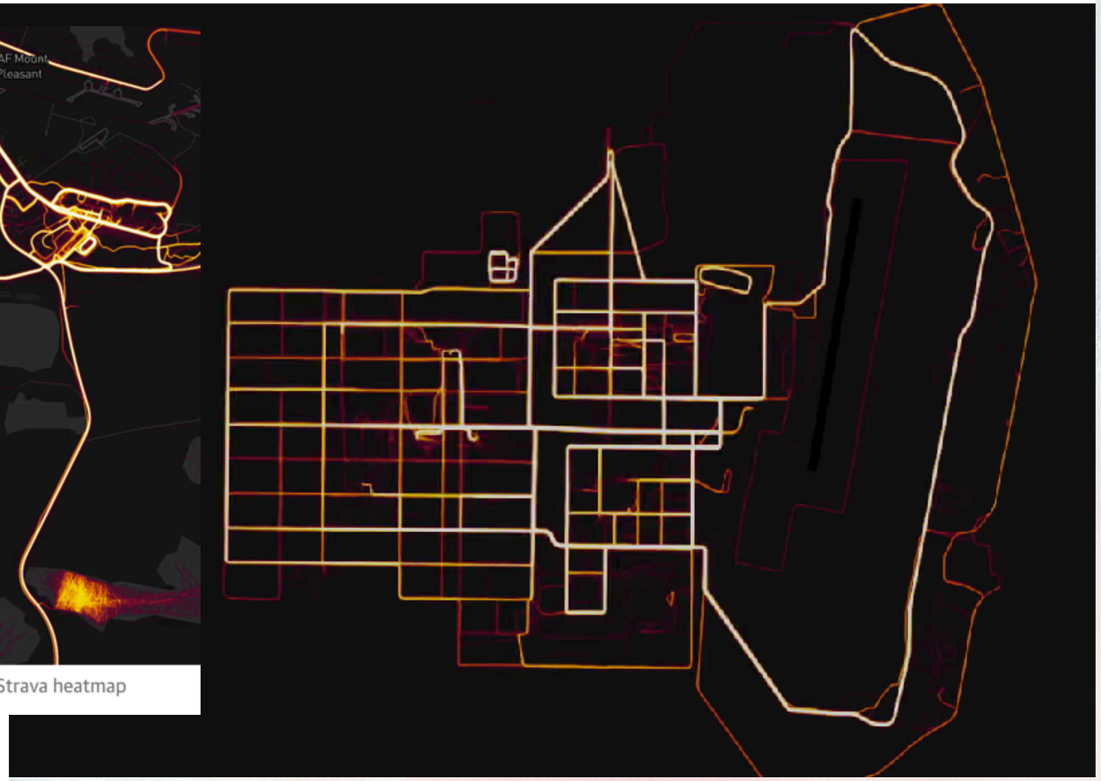
Latest: Strava suggests military users 'opt out' of heatmap as row deepens



Helmand province in Afghanistan. Photograph: Strava heatmap



RAF Mount Pleasant in the Falkland Islands. Photograph: Strava heatmap



NYC taxi data

- 173m individual taxi trips
- 20GB of uncompressed data

https://chriswhong.com/open-data/foil_nyc_taxi/

trip_data_1.csv

```
1  medallion, hack_license, vendor_id, rate_code, store_and_fwd_flag,
   pickup_datetime, dropoff_datetime, passenger_count, trip_time_in_secs,
   trip_distance, pickup_longitude, pickup_latitude, dropoff_longitude,
   dropoff_latitude
2  89D227B655E5C82AECF13C3F540D4CF4,BA96DE419E711691B9445D6A6307C170,CMT,1,N,2013-01-
   01 15:11:48,2013-01-01 15:18:10,4,382,1.
   00,-73.978165,40.757977,-73.989838,40.751171
3  0BD7C8F5BA12B88E0B67BED28BEA73D8,9FD8F69F0804BDB5549F40E9DA1BE472,CMT,1,N,2013-01-
   06 00:18:35,2013-01-06 00:22:54,1,259,1.50,-74.006683,40.731781,-73.994499,40.75066
4  0BD7C8F5BA12B88E0B67BED28BEA73D8,9FD8F69F0804BDB5549F40E9DA1BE472,CMT,1,N,2013-01-
   05 18:49:41,2013-01-05 18:54:23,1,282,1.10,-74.004707,40.73777,-74.009834,40.726002
5  DFD2202EE08F7A8DC9A57B02ACB81FE2,51EE87E3205C985EF8431D850C786310,CMT,1,N,2013-01-
   07 23:54:15,2013-01-07 23:58:20,2,244,.70,-73.974602,40.759945,-73.984734,40.759388
6  DFD2202EE08F7A8DC9A57B02ACB81FE2,51EE87E3205C985EF8431D850C786310,CMT,1,N,2013-01-
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7  20D9ECB2CA0767CF7A01564DF2844A3E,598CCE5B9C1918568DEE71F43CF26CD2,CMT,1,N,2013-01-
   07 15:27:48,2013-01-07
   15:38:37,1,648,1.70,-73.966743,40.764252,-73.983322,40.743763
8  496644932DF3932605C22C7926FF0FE0,513189AD756FF14FE670D10B92FAF04C,CMT,1,N,2013-01-
```

Data doesn’t need names to identify people

Re-identification	Mapping data back to a real person
Deanonymization	Breaking an anonymity claim
Linkage attack	Joining with auxiliary datasets
Singling-out	Isolating one individual in a dataset
Inference attack	Learning hidden attributes
Reconstruction	Rebuilding sensitive data
Linkability	Tracking the same person across records
Semantic leakage	Identity revealed via meaning/content
Trajectory attack	Identifying people via movement



Level 3: advanced techniques



k-anonymity



For every record, there must be **at least k people** with the same visible attributes

Methods

- Suppression
- Generalization

Age: 27
ZIP: 12345
Gender: F
Religion: Hindu

→

Age: 25 - 30
ZIP: 123**
Gender: F
Religion: *

Bucketing



Bucketing replaces exact values with ranges.

Age → 20–29
Salary → 50k–75k
Location → city instead of GPS
Time → hour instead of second

Adding noise



Adding tiny random changes to data to make individual values unreliable, but aggregates stay the same.

Name: Tom
Salary: 85000

→

Name: Tom
Salary: 87700

Methods

- Laplace
- Gaussian

Synthetic data generation



Strictly speaking data generation avoids the privacy problem entirely rather than solving it

Pros

- No direct identifiers
- No linkage attacks, no inference about real people
- Useful for sharing & development
- Often **far safer** than “anonymized” real data

Cons

- May reproduce near-identical individuals
- Poor for behavioral realism
- Deterministic and random

Make privacy a property of the stream

Privacy toolbox



Prevent sensitive data from entering



- Schemas
- Data contracts
- Data quality rules
- PII tags and field classification

Reduce data usefulness to attackers

- Aggregation
- Bucketing
- Synthetic data
- Noise or differential privacy

Protect sensitive fields



- Masking
- Tokenization
- CSFLE for selected fields
- CSPE for full-payload encryption

Know and prove what happened

- Stream Catalog
- Stream Lineage
- Audit logs

Restrict access



- Service accounts
- RBAC / Kafka ACLs
- Component-level permissions
- OAuth/OIDC or mTLS
- Private networking and customer-managed keys

Ask three questions

- Do we need this data?
- Who needs to see it?
- Can we prove where it went?